



Antibiotic-Carrying Micromotors

A group of nanoengineers have figured out how to transport antibiotics directly to the site of infection using micromotors. Learn more: <http://dgit.in/Micromotors>



A \$35,000 electric off-roader

Utah startup, Nikola Powersports has a brand new all-electric \$35,000 off-roader that has more torque than a tank. <http://dgit.in/electricTank>

Biological Age:

How old are you really?

We've been calculating our age in days and years since forever. We even use that same value to provide insurance, pension and more. What if we're wrong?

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Unlocking the secrets of ageing, or mortality is a long overdue achievement that humanity has always sought after. Generally, when you look at a person, you can tell in which broad age category they fall – child, middle aged, old etc. Knowing the exact date of birth is even better – you can simply calculate a person's age. But what if that is not the right way to tell how old someone is and we've been doing it wrong all along? What if you've been considering yourself an 18-year-old youngster when you are actually a 40-something person? Sounds absurd? This is where the concept of biological age comes in.

Have you ever noticed how two individuals in their 60s could have different levels of cognitive abilities? One might be as healthy and hearty as a middle aged person while the other might need constant supervision and medical attention. Scientists have pointed out this difference to explore the idea of biological age.

There are multiple methods that can be used to determine biological age. Generally, all anti-aging studies and methods look for genes that are associated with disease or are responsible for the process of aging. But one process, called RNA-profiling, is used to measure gene expression in tissue samples. Specifically, it looks at the activation of 150 genes in multiple tissue samples that showcase a good health status at a particular age. These parameters are then used to determine how well a person is ageing compared to other people born in the same year.

This score is an indicator of quite a few things – one of them being the cognitive decline in low scoring individuals. In the study which established this score, age related diseases like dementia or osteoporosis were better indicated by biological age rather than chronological age. Take, for instance, Alzheimer's disease in older individuals – the lower scores in individuals with the disease reflect a faster – or worse – ageing process than their counterparts with the same chronological age.

DOES IT WORK?

While the parameter discussed above might prove to be effective in determining high-risk of Alzheimer's disease in people, it is also true that no single measurement is enough to encompass the entire biological complexity behind ageing. Another testing method, GlycanAge Test uses Immunoglobulin G (IgG) content in blood. This protein, linked to im-

mune response, is modified by sugar molecules, known as glycans, over time. Studying the patterns of these molecules has indicated that their pattern changes with age and takes into account the lifestyle choices that you've made. This particular parameter is useful in determining how prone you are to lifestyle-related diseases like heart disorders, lung failure etc.

WHY IS THIS IMPORTANT?

Age, in today's society, determines a few important things – pension, retirement, insurance premium, loans etc. Using a blanket metric – chronological age – for people to determine their



Biological age relies on changes in DNA markers.

eligibility and the terms they get for the services mentioned earlier might be a flawed method that is doing injustice to a lot of healthy, fit individuals. On the other hand, being able to determine the chances of a particular disease showing up in a person based on their genetic data could be really important. It could give individuals the crucial time they need to prepare for the disease or even take measures to avoid it entirely. Regardless of the possibilities, determining your age is not as straightforward as you think. **4**

Source: <http://dgit.in/ageisjustano>